Aligned but Inert: Concept Bottlenecks in Dense Prediction, and How to Make Them Causal

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Abstract

Concept bottleneck models promise intrinsic interpretability: route the prediction through a human-readable concept layer, and the concept activation is *the explanation. We build one for dense prediction—a routing bottleneck that assigns every encoder token to one of K clinically labeled experts—and train it for 3D brain-tumor segmentation. By every measure this literature reports, it works: routing tracks the clinical concepts at $\mathrm{CAS}_{\mathrm{fg}}$ up to 0.96 and separates necrosis, edema and enhancing tumor at AUROC 0.88, against 0.74 for the strongest post-hoc attribution. Yet the routing is* causally inert. *Forcing tokens through a different expert changes 1.1% of voxel labels, and zeroing the bottleneck outright costs at most 3.1 Dice—the decoder reconstructs the segmentation without ever consulting the concept layer. Alignment metrics cannot detect this, because alignment is compatible with a bottleneck the model never consults, and the alignment loss trains precisely what those metrics score. The cause is not degenerate experts, which compute genuinely distinct transformations, but two leaks that encoder–decoder architectures invite: a residual path* through *the module and skip connections* around *it. We distill the diagnosis into three cheap tests—causal intervention, bottleneck ablation, and a probe control—that expose the failure without retraining, and then repair it.* Removing *the decoder's alternatives does not work; the optimizer compensates. Making the routing an* additive term of the output *does: causal control rises from 0.011 to* $\mathbf{0.636}$ $[0.621, 0.651]$ *label-flip under intervention ($p<10^{-28}$), the bottleneck turns from optional into indispensable (necrosis Dice $0.48 \to 0.01$ without it), and the price is ≈ 1 Dice point.* ***A concept layer is an explanation only if the model consults it, and showing that requires intervening, not correlating.***

1. Introduction

Deep segmentation models match expert radiologists on brain-tumor benchmarks, yet clinical deployment lags because their decisions cannot be verifiably explained. The dominant remedy is post-hoc attribution [18, 25, 27], which produces a saliency map *after* the prediction and therefore approximates the mechanism rather than reporting it. Rudin [24] argues for the alternative: build models whose decisions are interpretable by construction. Concept Bottleneck Models (CBMs) [15] realize this for classification by forcing the prediction through a human-readable concept layer, so that the concept activation *is* the explanation.

Extending that idea to dense prediction is attractive and—as we show—treacherous. We build the natural extension: replace the encoder bottleneck of a segmentation network with a router that assigns every token to one of K clinically labeled experts {Background, necrotic core (NCR), edema, enhancing tumor (ET)}. The routing distribution is a per-region, concept-labeled map computed in one forward pass with no post-hoc step. It looks exactly like an intrinsic explanation should.

It aligns beautifully, and it does nothing. Trained end-to-end, the routing tracks the clinical concepts closely ($\mathrm{CAS}_{\mathrm{fg}}$ up to 0.96) and discriminates them far better than any post-hoc method (0.88 vs. 0.74 for the strongest baseline). These are the metrics this literature reports, and by both the module is a success. But when we *intervene*—force the tokens the router assigned to one concept through a different concept's expert, changing nothing else—the prediction does not move. Only 1.1% of affected voxels change label, and zeroing the entire bottleneck costs at most 3.1 Dice.

The failure is invisible to alignment metrics *by construction*. $\mathrm{CAS}_{\mathrm{fg}}$ measures correlation between the routing and the ground truth, and correlation is perfectly compatible with a bottleneck the model never consults. Worse, the alignment loss trains precisely what $\mathrm{CAS}_{\mathrm{fg}}$ scores, so a high value confirms that the router learned its supervision—not that the network uses it.

Why it happens. We rule out the obvious explanation: the experts have *not* collapsed to a shared function—they displace tokens by 59–74% of their norm and differ pairwise by 0.93. The cause is architectural: two independent leaks that encoder–decoder designs invite. The expert computes $h + c_k(h)$, so the residual carries the full feature vector to the decoder whichever expert fires; routing modulates the representation but never gates it. And skip connections carry encoder features straight to the output, so the decoder can decide what tissue it sees without consulting the routing. Either leak alone suffices, and neither is visible in $\mathrm{CAS}_{\mathrm{fg}}$.

A control that reframes the baseline. A linear probe on an encoder trained with *no* concept supervision already separates the concepts at 0.81—above every post-hoc method we test—so discriminability gains over saliency baselines are far weaker evidence for a concept module than the literature treats them as.

Closing the leaks. Attempts to force dependence by *removing* the decoder's alternatives fail: a scalar gate on the skip connections is compensated away during training, and removing the expert residual backfires, collapsing pairwise expert divergence from 0.93 to 0.16 because every expert must then independently reconstruct the full signal. Making the routing an *additive term of the output* succeeds, because causal influence becomes arithmetic rather than something the optimizer can route around. Causal control rises from 0.011 to 0.636, removing the routing term now collapses the prediction (necrosis Dice $0.48 \to 0.01$), and the price is $\approx$1 Dice point.

Contributions.

1. **A diagnostic protocol for concept bottlenecks** (Sec. 4): causal intervention, bottleneck ablation, and a linear-probe control. Three cheap tests that separate a bottleneck the model uses from one it ignores. Alignment metrics cannot.
2. **Evidence that the natural design fails it** (Sec. 5.3). A bottleneck with $\mathrm{CAS}_{\mathrm{fg}}$ up to 0.96 and best-in-class concept-discriminability is causally inert, and we localize the cause to two specific architectural leaks.
3. **A repair and its price** (Sec. 5.6). Additive routing makes the bottleneck causally decisive—0.636 $[0.621, 0.651]$ label-flip under intervention against 0.011 $[0.009, 0.014]$, $p < 10^{-28}$—for $\approx$1 Dice point, with alignment and calibration slightly improved.

2. Related Work

Post-hoc attribution. Grad-CAM [25], integrated gradients [27], SHAP [18] and attention rollout [1] reconstruct a saliency map after the prediction. Their reliability has itself been questioned: popular methods can fail basic sanity checks [2], and the analogous audit found attention weights a poor guide to what a model used [13]. We apply that tradition to intrinsic rather than post-hoc explanation. On dense prediction the extension proceeds through heuristics, and the output is an input-*sensitivity* map rather than a concept assignment. We use four such methods as baselines under an identical protocol, scored by the deletion and insertion curves introduced with RISE [21]. Their central limitation for our purposes is structural rather than empirical: a saliency map is a readout, so there is nothing in it to intervene on. The causal test in Sec. 4 simply cannot be run on them.

Intrinsic interpretability and concept bottlenecks. CBMs [15] and ProtoPNet [6] make the explanation part of the computation, for scalar-output classification. Concepts have also been probed in networks not built around them [4, 14]; those methods measure alignment between an internal signal and a concept, precisely the quantity we show is insufficient. Neither addresses dense prediction, where the explanation must be per-voxel. Critically for this paper, the standard validation in this line is *concept accuracy or alignment*—does the concept layer predict the annotated concept—and not whether the downstream head consults it. That is the gap this paper measures. Reported failure modes for CBMs concern what the concept vector *encodes*—whether it carries task information beyond the named concepts. Ours is complementary and more basic: the concept vector may encode the concepts perfectly and still not be read, because the information bypasses the bottleneck through the *architecture* rather than through the concept encoding.

Mixture-of-experts routing. Sparse MoE [8, 22, 26] introduced routing for conditional computation. There, routing is causally decisive because the selected expert's output *is* the layer output. Our results show that transplanting the mechanism into a skip-connected encoder–decoder silently voids that property—which is exactly the assumption an interpretability claim would rest on. We inherited the intuition without checking that the conditions supporting it still held.

3. Concept Routing at the Bottleneck

3.1. Router, experts, dispatch

Fig. 1 gives the module and the two leaks at a glance. Let $h_t \in \mathbb{R}^D$ be the bottleneck token at spatial position t. A

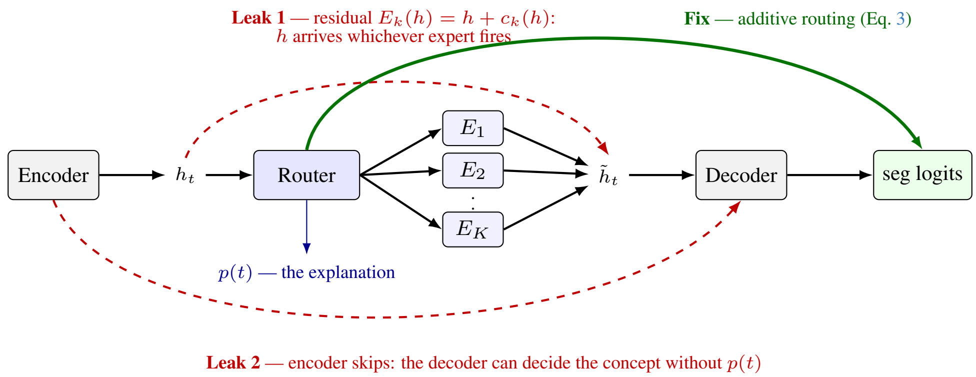

Figure 1. **Concept routing, and the two paths that make it inert.** The router assigns each bottleneck token h_t to one of K clinically labeled experts; the routing distribution $p(t)$ is read from the forward pass and is the explanation. The design is intrinsic only if $\tilde{h}_t$ is the decoder's *sole* source of concept information, and two standard choices break that: the expert residual delivers h intact whichever expert fires, and encoder skips let the decoder decide the concept without consulting $p(t)$. Either alone makes the routing inert, and neither is visible in any alignment metric. The fix adds the routing logits to the output rather than removing the decoder's alternatives, which the optimizer absorbs (Sec. 3.3).

router produces

$$p(t) = \mathrm{softmax}\big(W h_t/\tau + \beta\,\mathrm{sim}(h_t, \Pi)\big) \in \Delta^{K-1}, \quad (1)$$

where Π are K learnable concept prototypes and τ an annealed temperature. Each concept owns an expert E_k. Training uses soft dispatch, inference hard dispatch:

$$\tilde{h}_t = \sum_k p_k(t) E_k(h_t) \quad \text{(soft)}, \qquad \tilde{h}_t = E_{k^*(t)}(h_t) \quad \text{(hard)}, \quad (2)$$

where $k^*(t) = \arg\max_k p_k(t)$ selects the highest-probability expert. $p(t)$ is the explanation: a per-region distribution over clinical concepts, read directly from the forward pass. We train with $\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_1 \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{div}} + \lambda_3 \mathcal{L}_{\text{ent}} + \lambda_4 \mathcal{L}_{\text{bnd}}$, where $\mathcal{L}_{\text{align}}$ is a foreground-only focal BCE pushing $p_k(t)$ toward the ground-truth concept indicator, under a three-phase curriculum (warm-up / alignment / refinement) that anneals τ from 2.0 to 0.5.

Note already that $\mathcal{L}_{\text{div}}$ acts on routing *vectors*, not on expert *functions*: nothing in the objective forces $E_a \neq E_b$. We therefore verify functional divergence empirically rather than assume it (Sec. 5.3).

3.2. Two leaks

Eq. 2 yields an intrinsic explanation *only if* $\tilde{h}_t$ is the decoder's sole source of concept information. Two standard design choices break that.

Residual (through the module). The natural expert is a pre-norm residual MLP, $E_k(h) = h + c_k(h)$, whose residual stabilizes early training. It also means the decoder receives h intact regardless of k^*: routing reweights a signal that arrives anyway.

Skips (around the module). Encoder–decoder segmentation networks connect encoder stages directly to matching decoder stages [23], a design so standard that it is the default in the strongest current medical segmentation systems [12]. Every such connection is a path from input to output that does not pass the bottleneck. A related placement issue is easy to miss: in a hierarchical transformer encoder the "bottleneck" is ambiguous, and inserting the module at an intermediate stage leaves *deeper*, more semantic stages flowing around it. We report this because our own first design made exactly that error.

3.3. Three ways to close them

We expose all three as switches, and evaluate each.

(i) Gated expert. $E_k(h) = c_k(h)$, so the expert transform is the only path. The standard near-zero initialization of c_k's output layer must be relaxed here, since c_k is now the signal rather than a correction.

(ii) Skip attenuation. A scalar α scales the bypassing skips, sweeping from $\alpha{=}1$ (standard) to $\alpha{=}0$ (bottleneck is the only route). Skips carry spatial detail as well as semantics, so α trades boundary quality against causal control.

(iii) Additive routing. Rather than removing the decoder's alternatives, make the routing a term of the output:

$$\text{seg\_logits} = \mathrm{up}\big(\ell(t)\big) + \lambda \cdot \text{decoder\_out}, \quad (3)$$

where $\ell(t)$ are the router logits upsampled to voxel resolution and λ scales a bounded refinement. The routing supplies the coarse per-concept prediction; the decoder keeps every skip and contributes detail—which is what skips are good at.

Options (i) and (ii) are *subtractive*: they fight an architecture that segments perfectly well without its bottleneck.

Option (iii) is *additive*, and the distinction turns out to be decisive. Under Eq. 3 causal influence is arithmetic: at inference the refinement is a function of decoder features, so intervening on the routing shifts the output by exactly $\mathrm{up}(\Delta\ell)$. No weight configuration undoes an additive term after the fact.

The corresponding failure mode is magnitude rather than mechanism: the network can shrink the routing term *relative* to the refinement, leaving the effect arithmetically real and numerically negligible, since $\mathcal{L}_{\text{align}}$ pins the routing's direction but not its scale. We therefore monitor $\rho = \|\lambda \cdot \text{refine}\| / \|\mathrm{up}(\ell)\|$ throughout training rather than assuming it stays small.

3.4. Intervening on a two-path module

One implementation detail is essential and easy to get wrong. Under Eq. 3 the routing reaches the output through *two* paths: the expert features the decoder consumes, and the additive term. An intervention overriding only the dispatch leaves the additive path reading the router's original logits, so it silently does nothing through the path that matters. We therefore expose an *effective* logit tensor reflecting the forced decision, which the model consumes; the reported explanation stays the router's actual belief, so an intervention cannot contaminate the map we publish.

The hardening must also be applied uniformly. If only the overridden tokens become one-hot, an identity intervention $a \to a$ swaps soft logits for a hard assignment and moves the prediction though nothing was re-routed—here by 14.3 logits—destroying the diagonal control that licenses reading the off-diagonal cells. Applying it to every token makes baseline and intervention commensurable and restores $a \to a$ to a bit-exact no-op. We verified both before trusting any number in Sec. 5.6.

The same care applies to ablation. Patching only the decoder's bottleneck input leaves the additive term intact and measures a 0.09 logit change where the true effect is 7.0; a study using that measurement would conclude the bottleneck is unnecessary and reject a working design.

4. A Diagnostic Protocol

Alignment is necessary but not sufficient. We propose three tests, all cheap, that together establish whether a concept bottleneck is load-bearing.

Definitions. $\mathrm{CAS}_{\text{fg}}(k) = \mathrm{Pearson}\big(p_k(t), M_k(t)\big)$ over foreground tokens measures routing/ground-truth *alignment*, where M_k is the ground-truth concept indicator. Concept-discriminability is the AUROC of p_k as a classifier for concept k *among the other tumor voxels*—a harder question than localization, which only asks tumor vs. brain.

Test 1: causal intervention. Replace $k^*(t)$ with a chosen b for the tokens assigned to a, changing nothing else, and re-run the decoder. Report the change in the posterior of class b over exactly those voxels, the fraction of their labels that flip to b, the change in *off-target* classes (specificity), and the change *outside* the intervened region (locality).

Three details make this a control rather than a demonstration. The baseline must run through the same hard-dispatch path, so the identity intervention $a \to a$ is a bit-exact no-op and any off-diagonal effect is attributable to re-routing alone. When a model consumes routing through more than one path—as Eq. 3 does—the intervention must move *all* of them, or it silently does nothing through the one it missed. And it should be restricted to the tumor neighborhood: re-routing every token of a concept repaints the whole brain, showing that the output follows the routing everywhere rather than concept control where it is clinically meaningful (Sec. 5.6). We report both scopes and quote the restricted one.

This test is impossible for post-hoc attribution. There is nothing in a saliency map to intervene on, because the map is a readout: editing it changes nothing about the model.

Test 2: bottleneck ablation. Replace the module output with the raw encoder features (*bypass*) or with zeros (*zero*), and measure the Dice change. If destroying the bottleneck barely hurts, the decision does not live there, and no amount of concept alignment will make it interpretable. When the module reaches the output through two paths, each must be ablated separately: patching only the decoder's bottleneck input leaves an additive routing term intact and wrongly reports the bottleneck as unnecessary.

Test 3: linear-probe control. Fit multinomial logistic regression on the features *entering* the module and score it exactly as the router is scored. This asks whether the concept structure is produced by the concept machinery or merely read off features that already contained it. Run it also on a model trained without concept supervision, for the strong version of the control.

5. Experiments

5.1. Setup

BraTS 2024 glioma [3, 19]: four co-registered MRI modalities, labels remapped to {0:Background, 1:NCR, 2:Edema, 3:ET}, patient-wise $75/12.5/12.5$ split (n_{test}=168), z-scored per modality, 128^3 inputs, 80 epochs of AdamW (10^{-4}, cosine), mixed precision. The bottleneck is 192×16^3. Dice is reported two ways, labeled throughout: over the nested BraTS regions—whole tumor (WT), tumor core (TC), enhancing tumor (ET)—and *per class* (NCR, edema, ET). The two differ slightly on ET, since the regions are

Table 1. The natural design by standard metrics (BraTS test, n=168). Concept-discrim. is concept k vs. the *other* tumor voxels; 0.5 is chance.

	Dice WT/TC/ET	CAS_{fg} N/E/T	Concept-discrim.
NATURAL	0.93/0.88/0.87	0.80/0.96/0.95	**0.883**
Occlusion	—	—	0.741
Gradient	—	—	0.601
Int. grad.	—	—	0.583
Grad-CAM	—	—	0.521

nested and the classes are not. We report a SegResNet [20] backbone throughout for comparability across configurations; a Swin transformer with a UNETR decoder [10] reproduces every qualitative finding. All models are built on MONAI [5]. Three configurations recur:

NATURAL — residual expert, skips intact: the design a reader would build.

GATED — subtractive repair: gated expert, α=0.5.

DIRECT — additive routing, Eq. 3, λ=1.

Provenance. We diagnosed the failure first on Swin and replicated it on SegResNet, which agree on every qualitative finding (Sec. 5.8). Every headline number, and all of GATED and DIRECT, is SegResNet at n=168. Three NATURAL diagnostics ran on Swin only—the intervention pilot (n=5), expert divergence, and the linear probe—and carry a † wherever they appear.

5.2. The natural design aligns

Table 1 reports the standard picture. Segmentation is competitive, routing tracks the concepts closely, and on concept-discriminability NATURAL reaches 0.88 against 0.74 for the strongest post-hoc method, with integrated gradients and Grad-CAM near chance. By every measure this literature reports, the module works.

Removing $\mathcal{L}_{\text{align}}$ collapses CAS_{fg} to $-0.01/0.32/0.09$ while Dice barely moves (0.905 vs. 0.927 WT). We read this at the time as evidence that alignment is *learned* rather than incidental. It is—but the same number says something we missed: the segmentation does not need the alignment.

5.3. . . . and is inert

Intervention. Re-routing moves almost nothing. Averaged over concept pairs, 1.1% of re-routed voxels change label (Table 2, GATED row; NATURAL gives 0.1% on a smaller Swin pilot†). The identity diagonal is exactly 0, so this is a property of the model rather than of the instrumentation.

The experts are not degenerate. The obvious explanation is false. Experts displace tokens by 0.59–0.74 of their norm and differ pairwise by 0.93 (relative L_2, Swin†). The routing selects genuinely different transformations; they simply do not reach the output.

Bottleneck ablation. They do not reach the output because the output does not need them. Replacing the module with raw features or with *zeros* costs almost nothing: on SegResNet the worst concept loses 3.1 Dice (necrosis $0.514 \to 0.483$) and edema loses 0.6; on Swin†, edema $0.879 \to 0.865$ and ET $0.862 \to 0.806$. A decoder that can segment without its bottleneck will ignore any concept structure placed there.

5.4. The probe control

A linear probe on the features entering the module scores 0.839 against the router's 0.872 (Swin†). Run on a model trained with *no* concept supervision, the probe still reaches **0.811** —above every post-hoc method in Table 1. The concept structure is largely a property of a competent segmentation encoder, not of the concept machinery.

5.5. The output is a stronger concept classifier than the routing

A second control tightens the same point. The network's own segmentation posterior—free, from the same forward pass—separates the concepts at 0.979, well above the routing's 0.883. It reaches 0.982 on the transformer backbone despite different pretraining and Dice, so the metric saturates for any competent segmenter: under this window it nearly restates segmentation accuracy, because ranking voxels is far easier than thresholding them. Matched to the router's 16^3 decision budget by average-pooling and re-upsampling through the identical operator, the posterior falls to 0.935, so roughly half the residual gap is spatial resolution rather than concept information.

We report this because omitting it would overstate our case. The posterior is not an explanation: it *is* the prediction, and restating the output explains nothing about how the output arose. But with the probe control (Sec. 5.4) it establishes that concept-discriminability cannot carry an interpretability claim alone—an unsupervised linear probe clears the post-hoc baselines, and the model's own output beats everything.

5.6. Repair

Subtractive repair fails, instructively. Attenuating the skips (α=0.5) leaves every metric within noise of NATURAL: Dice -0.002 WT, CAS_{fg} -0.001 on edema, and causal control still at 0.011. A scalar gate is compensable during training—normalization affine parameters can simply rescale to undo it—so at convergence it constrains nothing.

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Table 2. Causal control and its price (BraTS test, n=168). Label-flip is the mean off-diagonal fraction of re-routed voxels whose predicted label became the target concept, tumor-restricted and scope-matched; the identity diagonal is exactly 0 in every configuration. Subtractive repair (GATED) does not work; additive routing (DIRECT) does, for ≈1 Dice point. †NATURAL's label-flip is a Swin pilot (n=5); its remaining columns and both other rows are SegResNet at n=168.

	Repair	Label-flip ↑	95% CI	Dice WT/TC/ET	$\mathrm{CAS}_{\mathrm{fg}}$ N/E/T	routing ECE ↓
NATURAL	none	0.001†	(Swin pilot, n=5)	0.927/0.883/0.869	0.80/0.96/0.95	0.841
GATED	gated expert, α=0.5	0.011	[0.009, 0.014]	0.925/0.872/0.854	0.79/0.96/0.95	0.673
DIRECT	additive routing	**0.636**	[0.621, 0.651]	0.927/0.873/0.858	**0.81**/0.96/0.95	**0.681**

Table 3. Bottleneck necessity for DIRECT (*per-class* Dice, n=40; ET therefore differs slightly from the nested-region ET of Table 2). Each path is ablated separately; *zero_routing* is the direct-mode number. Ablating only the decoder's bottleneck input would leave the routing term intact and wrongly report the bottleneck as unnecessary.

	base	bypass	zero_expert	**zero_routing**
NCR	0.483	0.484	0.053	**0.006**
Edema	0.898	0.477	0.333	**0.418**
ET	0.852	0.597	0.428	**0.172**

Removing the expert residual is worse than ineffective: pairwise divergence *collapses* from 0.93 to 0.16. With $E_k(h) = h + c_k(h)$ each expert supplies only a concept-specific correction and is free to differ; with $E_k(h) = c_k(h)$ every expert must independently reconstruct whatever the decoder needs, so they converge on nearly the same map. Making the expert the sole path made the experts interchangeable.

Additive routing works. Table 2 gives the result. Causal control rises from 0.011 [0.009, 0.014] to **0.636** [0.621, 0.651], a paired difference of +0.625 [+0.609, +0.641] with Wilcoxon p=2.6 × 10^{-29} over 168 matched cases. Fig. 2 shows the same result as images.

Two independent measurements corroborate it. Removing only the routing term collapses the model (Table 3): NCR −99%, ET −80%, edema −53%, against ≤6% for zeroing an inert bottleneck entirely. And ρ converged at 0.35, so the routing term dominates the logits roughly 3 : 1—the failure mode we instrumented for did not occur.

The price. Whole tumor is free (0.927 both); TC and ET cost ≈1 Dice point each. Alignment is marginally *better* than the inert baseline ($\mathrm{CAS}_{\mathrm{fg}}$ NCR 0.809 vs. 0.802), calibration improves substantially (expected calibration error, ECE, 0.841 → 0.681), and token-level AUROC rises on edema (0.638 → 0.900) and ET (0.622 → 0.839) while falling on NCR (0.937 → 0.783).

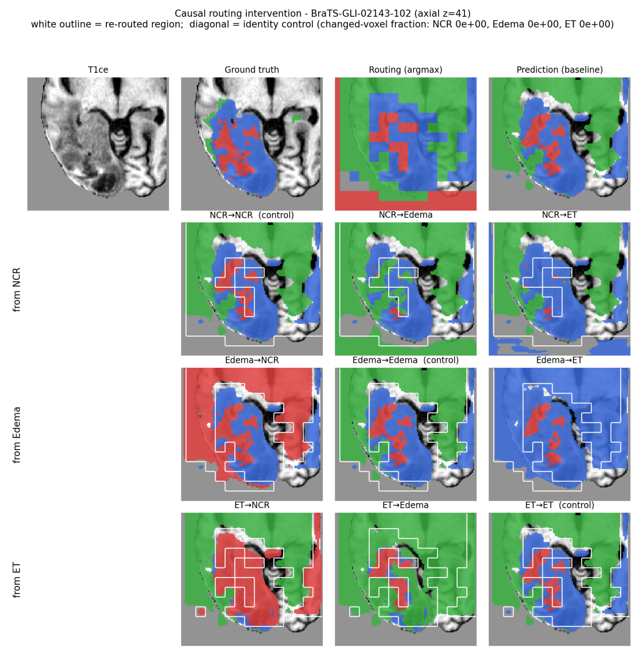

Figure 2. **Causal intervention, rendered.** Top: input, ground truth, routing argmax, baseline prediction. Grid: rows are the *source* concept, columns the *target*; each cell is the segmentation after forcing every token the router assigned to the row concept through the column concept's expert, with the re-routed region outlined in white. The diagonal is the identity control and is pixel-identical to the baseline (changed-voxel fraction 0 for all three), so the off-diagonal changes are attributable to re-routing alone.

Alignment emerges from the task. One consequence deserves emphasis. Because the routing logits are part of the output, $\mathcal{L}_{\mathrm{seg}}$ itself trains the routing to be concept-aligned. By epoch 10, with $\mathcal{L}_{\mathrm{align}}$ still switched off, $\mathrm{CAS}_{\mathrm{fg}}$ has reached 0.41/0.42 on edema and ET, against 0.16/0.15 for NATURAL. This directly answers the objection that $\mathrm{CAS}_{\mathrm{fg}}$ only measures what $\mathcal{L}_{\mathrm{align}}$ trains: here alignment appears without it.

The effect is specific and local. A large flip rate would mean little if the intervention perturbed everything, so we report two controls separating concept-specific control from

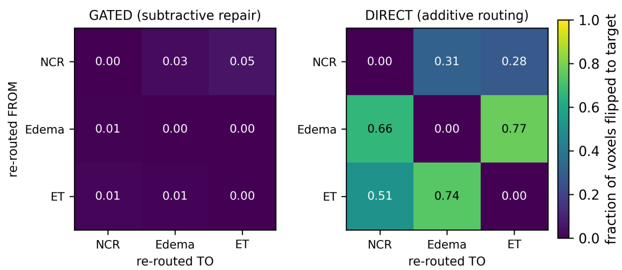

Figure 3. **Causal control, GATED vs. DIRECT**, on a shared color scale. Rows are the source concept, columns the target; each cell is the fraction of re-routed voxels whose predicted label became the target. Subtractive repair leaves the matrix uniformly dark; additive routing does not. Diagonals are the identity control and are exactly zero in both.

Table 4. Specificity and locality of the intervention (DIRECT, n=168, mean over off-diagonal concept pairs). Restricting the intervention to the tumor neighborhood does not merely lower the headline—it makes the effect *cleaner* on both controls.

	tumor-restricted	all tokens
target class ↑	**0.540**	0.798
source class ↓	−0.544	−0.514
off-target classes → 0	**0.079**	0.151
outside the region → 0	**0.005**	0.016

generic disturbance (Table 4). Off-target classes move by 0.079—an order of magnitude below the 0.636 target effect—and voxels *outside* the re-routed region by 0.005, while the source class falls by 0.544, mirroring the target's rise. The intervention does what it is supposed to do, where it is supposed to do it.

Scope matters, and the smaller number is the honest one. A router assigns most of the volume to background—here roughly a quarter of all tokens go to the edema expert, mostly normal tissue—so re-routing *every* token of a concept repaints the whole brain and reports 0.816 rather than 0.636. That figure is not wrong, but it mostly shows that the output follows the routing everywhere, which under Eq. 3 is close to tautological. Restricting to the tumor neighborhood measures concept control where it is clinically meaningful, and halves the off-target disturbance while cutting out-of-region leakage 3.5× (Table 4). We quote the restricted number throughout and report the scope with every measurement.

Where it is weakest, and why. The effect is not uniform, and the asymmetry is mechanistically informative rather than a mere caveat. Necrotic core is weak as a *source* (0.28–0.31) but works as a *target* (0.51–0.66): the model can be pushed into calling a region necrosis more easily than pushed off it. The source-class posteriors say why. Routing edema or ET tokens away drops their own class by 0.81 and 0.58, whereas routing necrosis away drops it by 0.23—the model was never confident about necrosis to begin with (isolated Dice 0.483, against 0.898 for edema). There is little confidence to remove, so the region is contested rather than reassigned and off-target classes absorb the difference (0.09–0.11 against 0.03–0.06 elsewhere). Causal control is therefore bounded by how decisively the model held the original concept—a property of the segmentation problem, not of the routing mechanism.

Table 5. The diagnostic protocol across configurations. Alignment separates none of them; the causal tests separate DIRECT from both. Ablation reports edema Dice with the module's contribution removed, relative to that configuration's own baseline. †NATURAL's three causal entries are Swin; GATED and DIRECT are SegResNet. Sec. 5.8 shows both backbones align and both are inert, so the separation the protocol reports is not a backbone effect.

	NATURAL	GATED	**DIRECT**
alignment — cannot separate them			
$\mathrm{CAS}_{\mathrm{fg}}$ (edema)	0.962	0.961	0.962
concept-discrim.	0.883	0.879	0.881
causal — separates DIRECT			
T1 intervention ↑	0.001†	0.011	**0.636**
T2 ablation, edema ↓	−1.6%†	−0.7%	**−53**%
T3 probe vs. router	0.839 vs. 0.872; 0.811 unsup.†		

5.7. The protocol, applied

Table 5 collects the three tests across the three configurations. NATURAL and GATED are indistinguishable on every causal test while differing on none of the alignment ones, which is the whole point: alignment cannot separate them, and intervention can.

5.8. Backbone transfer

The natural design behaves the same on both backbones. A Swin transformer [7, 17] with a UNETR decoder [11] reaches Dice 0.915/0.845/0.837, $\mathrm{CAS}_{\mathrm{fg}}$ 0.69/0.94/0.93 and discriminability 0.872; the SegResNet [20] CNN reaches 0.927/0.883/0.869, 0.80/0.96/0.95 and 0.883. They differ in pretraining, capacity and accuracy, yet both align well and both are inert, so the leaks are properties of the architecture family rather than of one backbone. The transformer shows the placement failure of Sec. 3.2 in its sharpest form: its module sits at stage 2 of 5, so the two deeper, more semantic encoder stages bypass it entirely.

6. Discussion and Limitations

What alignment metrics can and cannot show. $\mathrm{CAS}_{\mathrm{fg}}$ and concept-discriminability measure correlation between an internal signal and annotated concepts. A module can maximize both while contributing nothing to the prediction, and the alignment loss trains precisely what these metrics score. Any claim that a concept layer makes a model interpretable therefore requires an intervention. This lesson cost us: on the correlational evidence alone we had concluded the module worked, and only the causal test reversed that reading.

Additive beats subtractive. Both failed repairs share a shape: they force a decoder that segments well without its bottleneck to depend on it by removing alternatives, and the optimizer routed around both. Making the routing a term of the output instead makes causal influence arithmetic, which no weight configuration undoes after the fact. We expect this to generalize to other concept-bottleneck designs in dense prediction, though we have tested it only here.

Reported metrics that do not transfer. Two comparisons become invalid under Eq. 3, and we flag rather than tabulate them. Segmentation loss is not comparable across additive and non-additive configurations, since the larger logit scale lowers cross-entropy without improving the argmax. And GATED's low off-target figure (0.022 vs. DIRECT's 0.079) reflects its near-zero effect, not better precision—an inert module perturbs nothing, including what it should.

No controlled accuracy comparison. We did not train plain-backbone counterparts under our split, so we make *no* claim that the module is accuracy-neutral in absolute terms. The ≈ 1 Dice point in Table 2 is a *relative* cost against the same architecture with an inert bottleneck—the quantity the trade-off argument needs, but not the cost of adding the module to a plain network.

An unexplained interaction. Zeroing both CCR paths is *less* damaging than zeroing the routing term alone (edema 0.801 vs. 0.418, Table 3). The decoder is trained to add to a routing base; stripping the base while leaving the expert path appears to leave it miscalibrated, whereas zeroing both puts it in a cleaner regime. We report this because we cannot yet explain it.

Resolution, granularity, and scope. The bottleneck is 16^3, so the explanation is coarse, and necrotic core—tens of tokens per case—is the consistent weak spot. $K{=}4$ follows the BraTS protocol and forces transition-zone tokens into one concept. We evaluate one dataset and one anatomy; the leaks are properties of the architecture family rather than the data, but that expectation is untested. Routing entropy is a per-voxel uncertainty signal but poorly calibrated (ECE 0.68), and plain segmentation confidence predicts model error better than any routing-derived signal we tried (0.883 vs. 0.786), so we make no uncertainty claim and do not position routing entropy against MC-dropout [9] or deep ensembles [16].

What the explanation is, and is not. CCR explains at the level of clinical concepts—which expert processed each region, and with what confidence—not the router's internal reasoning. It is not spatially sharper than a saliency map, and the segmentation output remains a finer-grained concept map; what the routing adds is a low-dimensional, inspectable and *manipulable* account of the concept decision. We ran no reader study, so we make no claim that clinicians want this. What the result changes is the question the model can answer: a saliency map supports only "where did the model look", an aligned but inert bottleneck adds "what did the model call this region", and a *causal* bottleneck supports counterfactuals—"what if this region were edema?"—answered by the model itself. Before the repair that question was unanswerable in principle, because the output did not depend on the concept assignment.

Reproducibility. Configurations differ only in the switches of Sec. 3.3—backbone, losses, curriculum, optimizer, split and seed are shared—so the comparisons isolate the architectural change. Every number comes from the released scripts and committed per-case files, and the intervention diagonal is recomputed each run as a self-check.

Practical guidance. Three recommendations follow for anyone placing a concept layer in a dense-prediction network. Run the intervention before reporting alignment—it needs no retraining and was the only test separating a working design from a decorative one. Check where the bottleneck actually is: in a hierarchical encoder the deepest stage is not the one whose width matches the module. And prefer additive to subtractive coupling—both subtractive repairs we tried were absorbed away.

7. Conclusion

The obvious implementation of a concept bottleneck for dense prediction is beautifully aligned and causally inert: $\mathrm{CAS}_{\mathrm{fg}}$ up to 0.96, best-in-class discriminability, and a prediction that does not move when the routing is overwritten. The cause is two architectural leaks that alignment metrics cannot see, not degenerate experts. Additive routing raises causal control from 0.011 to 0.636 for ≈ 1 Dice point. **A concept layer is an explanation only if the model consults it, and demonstrating that requires intervening, not correlating.**

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